

Week 2 — Data Sources & Governance

How do I find the right biomedical data to answer my question?

BIOE 860 • Week 2

Learning objectives

- Classify biomedical data sources by origin and access model.
- Match a research question to an appropriate data source.
- Explain why governance, privacy, IRB review, and data-use agreements matter.
- Describe clinical data integration and common EHR harmonization problems.
- Navigate a data dictionary and translate a research question into variables.
- Compare UTHSC resources including Cerner Health Facts, rEDW, and TriNetX.

Start with the question — not the database

Case: Can we predict 30-day hospital readmission among adults with type 2 diabetes?

Population: Which patients count? Inpatient only? Adults? First admission or all admissions?

Exposure/predictors: demographics, diagnoses, medications, laboratory values, prior utilization?

Outcome: What exactly counts as a readmission? Same hospital? Any hospital? Within 30 days?

Time: What is the index encounter and what information is allowed before prediction?

Only after these decisions should we ask: **Which data source can support the study?**

A practical taxonomy of biomedical data sources

Public

Openly downloadable or broadly available; e.g., many CDC datasets.

Controlled / restricted

Access requires registration, training, DUA, proposal review, or a secure environment.

Institutional clinical

Local EHR/warehouse data; governance depends on institution, project, and identifiability.

Research-generated

Data collected specifically for a study, registry, trial, biobank, or laboratory experiment.

Public data: powerful, but not automatically 'easy'

- **Advantages:**
 - transparent access,
 - reproducible sharing
 - often strong documentation
 - useful for teaching and benchmarking.
- **Limitations:**
 - variables may be coarsened
 - rare conditions may be suppressed
 - linkage may be limited
 - sampling design can matter.

Public data: powerful, but not automatically 'easy'

Example: NHANES provides extensive public-use files, while some sensitive or disclosure-risk variables require controlled access through the NCHS Research Data Center.



Example: PhysioNet hosts both open-access datasets and controlled clinical datasets; access to resources such as MIMIC-IV requires credentialing, training, and a data use agreement.



Question to ask: Does the public-use version contain the variables and granularity needed for my hypothesis?

Restricted data: access is part of the research design

Possible requirements:

• user registration • institutional affiliation • human-subjects training • project proposal	• IRB determination • DUA (Data Use Agreements) • secure computing • output review
--	--

Restrictions protect participants and data providers, but they also affect timelines, collaboration, software choices, and reproducibility.

A project can be scientifically excellent and still be infeasible if access cannot be obtained in time.

Plan governance at the beginning—not after the analysis code is written.

Data Governance:

Who can do what with the data and under what conditions?

For example, imagine you want to use patient data from UTHSC data warehouse rEDW (Research Enterprise Data Warehouse). Governance covers questions such as:

- **Who can access it?** Students? Faculty? Outside collaborators?
- **Why can they access it?** Research, quality improvement, education?
- **What approval is required?** IRB determination, DUA, institutional authorization?
- **Where can the data be stored?** Laptop? Secure server? Approved cloud environment?
- **Can it be shared?** With collaborators? With another institution?
- **How long can it be retained?**
- **What happens when the project ends?**
- **Who is responsible if something goes wrong?**

Data Governance: Related but Different Concepts

Privacy

Rules and expectations about appropriate use of information.

Security

Technical/administrative safeguards that protect data.

IRB

Institutional review of human-subjects research or determination of status.

DUA

Agreement defining permitted uses, users, protections, and responsibilities.

De-identification

Removal/transformation of identifiers to reduce re-identification risk.

Consent

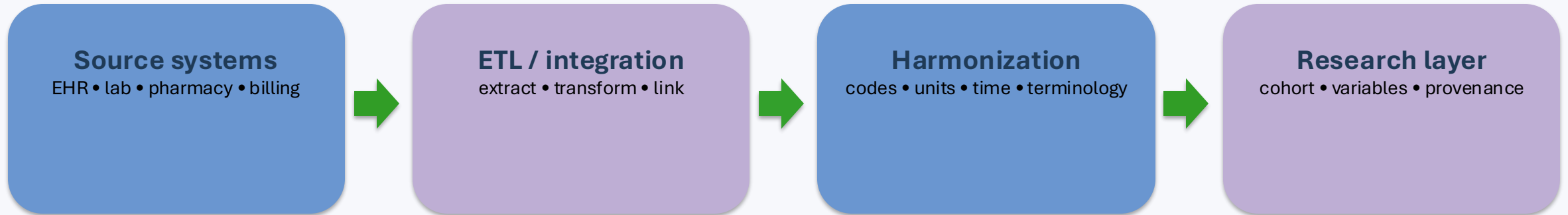
Participant authorization; scope may constrain future uses.

Clinical data were created for care, not for your analysis

- EHR fields reflect clinical workflows, billing, documentation habits, local conventions, and software configuration.
- Missing does not always mean 'not measured'; it may mean measured elsewhere, not ordered, not interfaced, or not captured in the extract.
- A diagnosis code can represent disease, rule-out evaluation, billing, history, or problem-list carry-forward.
- Medication orders, administrations, reconciliations, and prescriptions are different events.
- Clinical context is part of the data-generating process.

Clinical Data Integration

From fragmented clinical systems to a research-ready representation



- Integration is not concatenation. The same concept can use different identifiers, code systems, units, and timestamps across institutions.
- Preserve provenance: an analysis variable should be traceable to its source and transformation history.

Common integration problems to look for

- **Patient identity:** one person may have different identifiers across systems.
- **Encounter identity:** ED, observation, inpatient, and transfer episodes may be represented differently.
- **Terminology:** ICD-9 vs ICD-10; local lab codes vs LOINC; local drug names vs NDC/RxNorm.
- **Units:** mg/dL vs mmol/L; kg vs lb; local reference ranges.
- **Time:** order time, collection time, result time, admission time, discharge time—and time zones.
- **Provenance:** which hospital/system/version produced the value?

UTHSC clinical data ecosystem

Cerner Health Facts — rEDW — TriNetX Live

UTHSC clinical data ecosystem: Cerner Health Facts

- A large, de-identified, longitudinal electronic health record (EHR) database.
- Aggregates clinical data contributed by participating health systems across USA rather than representing only UTHSC patients.
- UTHSC's Center for Biomedical Informatics (CBMI) provides access for UTHSC researchers and affiliates.
- UTHSC describes the resource as containing almost 50 million patients and almost 300 million encounters.

≈50M
patients

≈300M
encounters

Cerner Health Facts:

What kinds of data are available?

Demographics

Encounters

Diagnoses

Procedures

Medications / prescriptions

Laboratory tests

**Care locations (clinic,
ED, ICU, etc.)**

**Hospital information &
billing**

Cerner Health Facts: What can researchers do with it?

- Describe real-world patterns of care across large patient populations.
- Study associations among diagnoses, treatments, laboratory results, procedures, and outcomes.
- Examine practice patterns and variation across hospitals, regions, or patient groups.
- Study drug or device use across diagnoses and major procedures.
- Develop and validate phenotypes, prediction models, and data-science workflows.
- Important: observational EHR data can support association and prediction, but causal questions require careful study design.

Cerner Health Facts at UTHSC

- Best fit: questions needing broad multi-institution clinical patterns rather than only local Tennessee care.
- Before choosing it: verify variable availability, temporal coverage, provenance, and the current access/governance requirements.

How do I access Cerner Health Facts at UTHSC?

- 1 **Define the research question** — Identify the cohort, variables, outcomes, and time window you need.
- 2 **Review the resource** — Check whether Health Facts contains the clinical elements required for the project.
- 3 **Request access through CBMI** — UTHSC's Cerner Health Facts page provides the database access-request route.
- 4 **Confirm regulatory requirements** — UTHSC's IRB lists Cerner Health Facts (USA) among certified de-identified repositories with NHSR status; confirm that your specific project qualifies.
- 5 **Work in the approved environment** — Follow CBMI/UTHSC requirements for data access, storage, analysis, and disclosure.

Health Facts vs. UTHSC's rEDW

Two useful but different clinical-data resources

Cerner Health Facts

- De-identified multi-institution EHR database
- Very large national-scale resource
- Useful for broad real-world clinical research
- Access provided through UTHSC CBMI

UTHSC Research Enterprise Data Warehouse (rEDW)

- Integrated clinical data from multiple UTHSC-affiliated/partner health systems
- Patient- and encounter-level research data
- Includes demographics, labs, pharmacy, diagnoses, procedures, notes and more
- UTHSC states IRB approval is required for rEDW research access

UTHSC rEDW and TriNetX

- rEDW integrates data from multiple clinical sources and provides a unified patient view.
- UTHSC lists demographics, laboratory results, pharmacy information, pathology, admission/discharge dates, ICD-9/10, CPT, discharge summaries, and progress notes among rEDW content.
- UTHSC states that IRB approval is required to use rEDW for research.
- **TriNetX** is cloud-based platform available through CBMI and can support cohort identification, feasibility, and protocol design using standardized aggregated clinical data.



TriNetX

- Practical workflow: explore feasibility → define cohort/variables → obtain approvals → request detailed data when needed.

Data dictionaries: the bridge between a question and a query

- A variable name is not a definition.
- For every candidate field ask: What does it mean? What is the unit? What is the coding system? At what level is it stored—patient, encounter, order, result?
- Check keys and relationships: PATIENT_ID, encounter/admission identifiers, order identifiers.
- Check time variables: which timestamp should define the clinical event?
- Check missingness and valid values before writing analysis code.



Running case: translate the question into data elements

Population: adults with type 2 diabetes admitted to hospital.

Index encounter: define an eligible inpatient admission and discharge.

Outcome: a new qualifying admission within 30 days.

Predictors: age, sex, prior utilization, diagnoses/comorbidity, medications, selected labs, length of stay.

Exclusions: decide how to handle death, transfers, planned admissions, incomplete follow-up, and duplicate encounters.

Now ask: Which source contains these elements with enough consistency to answer the question?

Source selection matrix

Question	Public survey	Cerner HF	rEDW	TriNetX
National prevalence estimate	Strong	Limited fit	Poor fit	Poor fit
Large EHR association study	Possible	Strong	Strong local/state	Feasibility
Local detailed patient-level study	Poor fit	Not local-specific	Strong	Screen first
Rapid cohort feasibility	Limited	Depends	Possible	Strong

Lab — Data dictionary detective

Task 1: For the readmission case, identify the minimum variables and tables required.

Task 2: For each variable, record definition, unit/coding system, table level, key identifier, and relevant timestamp.

Task 3: Identify at least three ambiguities that must be resolved before analysis.

Task 4: Decide whether Cerner Health Facts, rEDW, TriNetX, or a public dataset is the best starting point—and defend the choice.

Task 5: Sketch the access/governance steps you would expect before receiving patient-level data.

Three mini-scenarios: choose the source

- A. Estimate U.S. diabetes prevalence by age and sex using a nationally representative sample.
- B. Test whether inpatient laboratory patterns are associated with 30-day readmission across a very large de-identified EHR population.
- C. Identify whether enough eligible patients exist locally for a proposed UTHSC clinical study before writing the full protocol.

For each: choose a source, state the access level, name likely governance steps, and identify one important limitation.

Governance scenario: what would you do?

You receive a 'de-identified' clinical extract from a collaborator.

The file is on a shared cloud folder. There is no README, DUA, IRB determination, or documentation of how identifiers were removed.

The collaborator says: 'It's de-identified, so we're fine.'

Questions: What do you need to verify before using it? Who has authority to make the determination? Where should the data be stored? What documentation should exist?

Week 2 take-home messages

- The best dataset is the one that can validly answer your specific question—not necessarily the largest one.
- Access model and scientific quality are separate issues.
- Governance is part of study design and project planning.
- Clinical data integration changes the meaning and usability of EHR data; provenance matters.
- Read the data dictionary before writing the analysis.
- For UTHSC projects, CBMI resources such as Health Facts, rEDW, and TriNetX provide complementary routes to clinical data.

Selected resources

UTHSC Center for Biomedical Informatics (CBMI): clinical data resources and access <https://uthsc.edu/cbmi/>

UTHSC CBMI: TriNetX and Research Enterprise Data Warehouse (rEDW) <https://uthsc.edu/cbmi/services/trinetx.php>

UTHSC rEDW Data Dictionary <https://uthsc.edu/cbmi/data/documents/redw-data-dictionary-2022.pdf>

MIMIC (Medical Information Mart for Intensive Care): <https://mimic.mit.edu/>

Always verify current access and regulatory requirements before starting a project.